Kshitiz Gupta

Assignment 1: Linux Basic Commands

Note:In each and every terminal statement vboxuser@kshitizgupta means the entire line that is displayed in screenshot.

Section A:Navigation

1. Create a directory named linux_practice.
 vboxuser@kshitizgupta:~$ mkdir linux_practice

2. Go inside that directory.
 vboxuser@kshitizgupta:~$ cd linux_practice

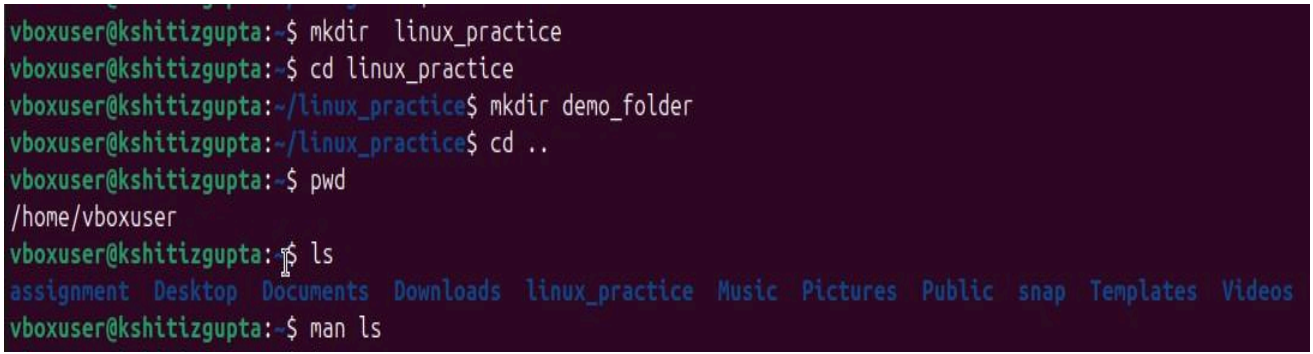
3. Create another directory called test_folder.
vboxuser@kshitizgupta:~/linux_practice$ mkdir demo_folder

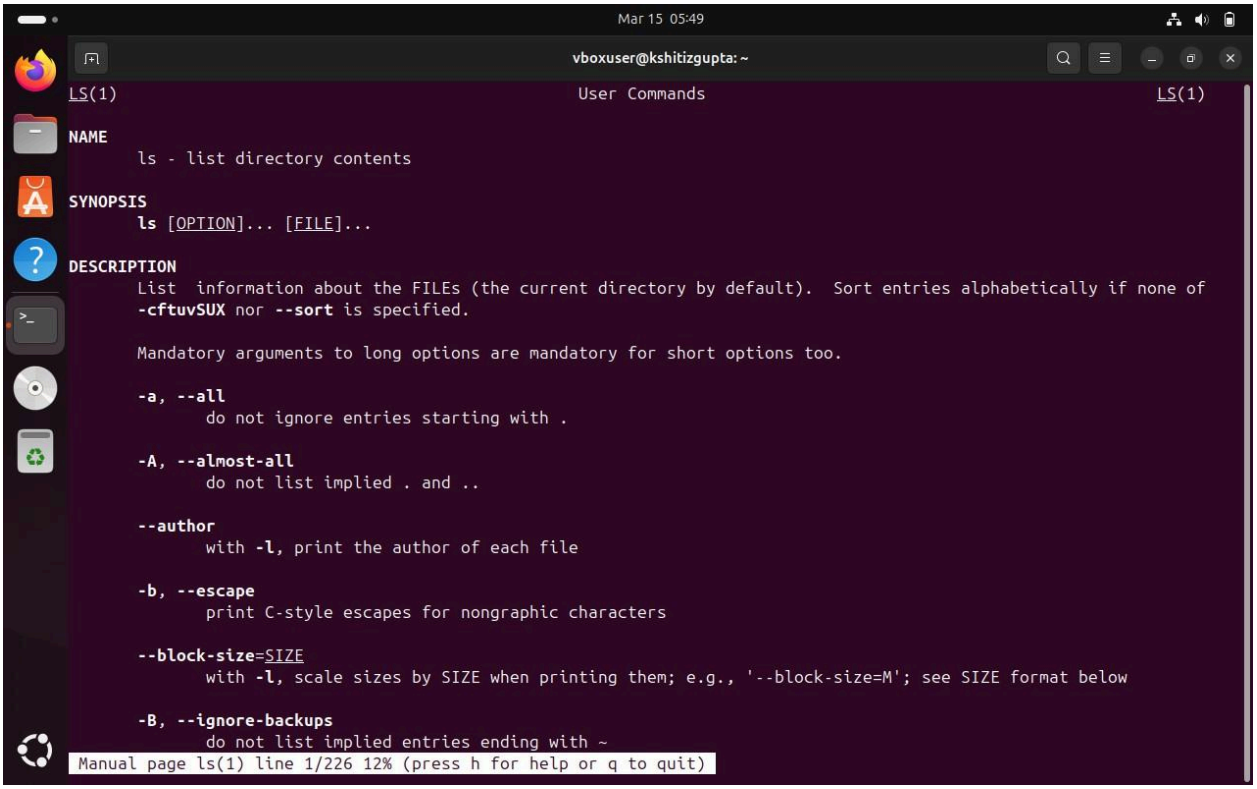
4. Move back to the parent directory.
vboxuser@kshitizgupta:~/linux_practice$ cd ..

5. Display your current working directory.
vboxuser@kshitizgupta:~$ pwd

6. List all files and directories.
 vboxuser@kshitizgupta:~$ ls

7. Find help for the ls command.
 vboxuser@kshitizgupta:~$ man ls





Section B: File & Directory Operations

1. Create a file named student.txt.

vboxuser@kshitizgupta:~$ touch student.txt

2. Create a directory called data.

vboxuser@kshitizgupta:~$ mkdir data

3. Copy student.txt inside data.

vboxuser@kshitizgupta:~$ cp student.txt data/

4. Rename student.txt to info.txt.

vboxuser@kshitizgupta:~$ mv student.txt info.txt

5. Delete the file info.txt.

vboxuser@kshitizgupta:~$ rm info.txt

6. Remove the data directory using:

○ rmdir

vboxuser@kshitizgupta:~$ rmdir data

○ rm -rf

vboxuser@kshitizgupta:~$ rm -rf data

```
vboxuser@kshitizgupta:/media/sf_Downloads$ touch student.txt
vboxuser@kshitizgupta:/media/sf_Downloads$ mkdir data
vboxuser@kshitizgupta:/media/sf_Downloads$ cp student.txt data/
```

```
vboxuser@kshitizgupta:/media/sf_Downloads$ mv student.txt info.txt
vboxuser@kshitizgupta:/media/sf_Downloads$ rm info.txt
vboxuser@kshitizgupta:/media/sf_Downloads$ rmdir data
rmdir: failed to remove 'data': Directory not empty
vboxuser@kshitizgupta:/media/sf_Downloads$ rm -rf data
```

Assignment 2: File Editing &Text Processing

1. Create a file called bio.txt.
vboxuser@kshitizgupta:~$ touch bio.txt

2. Add the following details using vi:
○ Your Name
○ Course Name
○ City
vboxuser@kshitizgupta:~$ vi bio.txt

3. Save the file properly.
ESC
:wq

4. Search for your name inside the file.
vboxuser@kshitizgupta:~$ grep "kshitiz" bio.txt

5. Delete all content from the file using shortcut commands. vboxuser@kshitizgupta:~$ vi bio.txt
ESC
:%d
:wq

6. Use grep to:
○ Search your name in bio.txt
vboxuser@kshitizgupta:~$ grep "kshitiz" bio.txt

○ Show lines that DO NOT contain your name. vboxuser@kshitizgupta:~$ grep -v "Kshitiz" bio.txt

VI editor:-

```
vboxuser@kshitizgupta: /media/sf_Downloads/linux_practice
Name: Kshitiz Gupta
Course: Linux Training
```

```
vboxuser@kshitizgupta:/media/sf_Downloads$ mkdir linux_practice
vboxuser@kshitizgupta:/media/sf_Downloads$ cd linux_practice
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch bio.txt
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi bio.txt
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ grep "kshitiz" bio.txt
 kshitiz gupta
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ grep -v "kshitiz" bio.txt
Name: Kshitiz Gupta
Course: Linux Training
```

Assignment 3: Linux Permissions

1. Create a file script.sh.
vboxuser@kshitizgupta:~$ touch
script.sh

2. Check its permission using ls -l.
vboxuser@kshitizgupta:~$ ls -l script.sh

3. Change permission to:
○ 755
vboxuser@kshitizgupta:~$ chmod 755 script.sh

○ 764
vboxuser@kshitizgupta:~$ chmod 764 script.sh

4. Explain what 764 means.

In Linux, every file and directory has **permissions** that control who can read, write, or execute the file. These permissions are assigned to **three groups of users**:

1. **Owner** – The person who created the file

2. **Group** – Users who belong to the same group

3. **Others** – All other users on the system

The permission **764** is divided into three parts:

Number	User Type	Permission
7	Owner	Read, Write, Execute
6	Group	Read, Write
4	Others	Read only

Explanation:

- **Owner (7)** → Can read, modify, and execute the file.
- **Group (6)** → Can read and modify the file but cannot execute it.
- **Others (4)** → Can only read the file but cannot change or execute it.

So, permission **764** allows the owner full access, the group limited access, and others only read access.

5. Give execute permission using symbolic method (chmod +x). vboxuser@kshitizgupta:~$ chmod +x script.sh

6. What is the meaning of r=4, w=2, x=1?
7. What does permission 777 mean? Is it safe? Why?

Permission 777 means:

User	Permission
Owner	rwx
Group	rwx
Others	rwx

Meaning:

Everyone can read, write and execute the file.

It is **not safe** because any user can modify or delete the file.

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls
bio.txt  script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls -l script.sh
-rwxrwx--- 1 root vboxsf 0 Mar 15 06:47 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod 764 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls -l script.sh
-rwxrwx--- 1 root vboxsf 0 Mar 15 06:47 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod +x script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls -l script.sh
-rwxrwx--- 1 root vboxsf 0 Mar 15 06:47 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod 777 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls -l script.sh
-rwxrwx--- 1 root vboxsf 0 Mar 15 06:47 script.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$
```

Assignment 4 : Linux Networking

Task-1 : Check Network Configuration

1. Check your system IP address.

vboxuser@kshitizgupta:~$ ip a

2. Write down about:

○ IP Address

IP Address stands for **Internet Protocol Address**.
It is a unique number assigned to each device connected to a network.

It allows computers to **communicate with each other over the internet or a local network**.

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f2:e6:87 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 81222sec preferred_lft 81222sec
    inet6 fd17:625c:f037:2:bc9e:360:d031:74a8/64 scope global temporary dynamic
       valid_lft 86380sec preferred_lft 14380sec
    inet6 fd17:625c:f037:2:a00:27ff:fef2:e687/64 scope global dynamic mngtmpaddr
       valid_lft 86380sec preferred_lft 14380sec
    inet6 fe80::a00:27ff:fef2:e687/64 scope link
       valid_lft forever preferred_lft forever
```

The command `ip a` is used in Linux to display network interfaces and IP addresses.
In my system, the active network interface is **enp0s3**, and the assigned IPv4 address is **10.0.2.15**.
This IP address is provided by the VirtualBox NAT network and is used for communication with other systems.

Task-2 : Test Network Reachability

3. Ping google.com
vboxuser@kshitizgupta:~$ ping google.com

4. Observe and write:

○ IP address resolved

IP address resolved

When you run ping, **google.com is converted into an IP address**.

Example:

142.250.183.14

This process is called **DNS resolution**.

○ Packet loss %

At the bottom of ping output you will see something like:
0% packet loss
Meaning:

- **0% packet loss** → network connection is working properly
- **High packet loss** → network problem

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ping google.com
PING google.com (142.251.223.206) 56(84) bytes of data.
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=1 ttl=255 time=209 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=2 ttl=255 time=32.2 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=3 ttl=255 time=31.1 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=4 ttl=255 time=31.5 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=5 ttl=255 time=30.4 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=6 ttl=255 time=32.0 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=7 ttl=255 time=32.2 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=8 ttl=255 time=32.6 ms
64 bytes from tzdela-as-in-f14.1e100.net (142.251.223.206): icmp_seq=9 ttl=255 time=30.0 ms
```

Task-3 : Check Port Connectivity

5. Check if port 80 is open on google.com using

telnet: vboxuser@kshitizgupta:~$ telnet google.com

80

6. What does it mean if:

7. What does it mean if:Download google.com homepage.

○ Connection

successful? It means:

- The server is reachable
- The port is open
- Network communication is working correctly

○ Connection

refused? It means:

- The server is reachable
- But the port is closed
- No service is running on that port

○ Connection timed out?

It means:

- The server cannot be reached
- Network or firewall may be blocking the connection

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ telnet google.com 80
Trying 142.250.182.206...
Connected to google.com.
Escape character is '^]'.
```

Task-4 : Download Data Using curl

8. Save it into a file.
vboxuser@kshitizgupta:~$ curl google.com -o google.html

9. Verify the file is created.

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ curl google.com -o google.html
Command 'curl' not found, but can be installed with:
sudo snap install curl  # version 8.19.0, or
sudo apt  install curl  # version 8.5.0-2ubuntu10.6
See 'snap info curl' for additional versions.
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ^C
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ sudo snap install curl
[sudo] password for vboxuser:
curl 8.19.0 from Yuzukosho (aoilinux) installed
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ curl google.com -o google.html

    Caution: You are using the Snap version of curl.
    Due to Snap's sandbox nature, this version has some limitations.
    For example, it may not be able to access hidden folders in your home directory
    or other restricted areas of the os.
```

vboxuser@kshitizgupta:~$ls

Assignment 5:Process & System Monitoring

vboxuser@kshitizgupta:~$ ps

Open top command and monitor CPU usage.
vboxuser@kshitizgupta:~$ top

Check disk usage.
vboxuser@kshitizgupta:~$ df -h

Check memory usage.
vboxuser@kshitizgupta:~$ free -h

Difference between ps and top ps:

Feature	`ps`	`top`
Type	Snapshot command	Interactive monitoring tool
Output	Shows a **static list** of processes at the moment you run it	Shows **real-time updating** process list
Interaction	Non-interactive	Interactive (you can sort, kill processes, etc.)
Refresh	Does **not refresh automatically**	**Auto refreshes** every few seconds

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           392M  1.6M  390M   1% /run
/dev/sda2        40G  6.1G   32G  17% /
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           392M  132K  392M   1% /run/user/1000
/dev/sr0        6.0G  6.0G     0 100% /media/vboxuser/Ubuntu 24.04.2 LTS amd64
Downloads       476G  360G  117G  76% /media/sf_Downloads
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       1.8Gi       712Mi        54Mi       1.6Gi       2.0Gi
Swap:             0B          0B          0B
```

Explanation:

The `ps` command provides a static list of running processes at the moment it is executed, while `top` provides a dynamic real-time view of system processes and resource usage such as CPU and memory.

The `df -h` command displays the disk space usage of file systems in a human-readable format. It shows total disk space, used space, available space, and the percentage of disk usage

Assignment 6: Shell Scripting

Task 1: Simple Script

Write a shell script that:

- PrintsWelcome to Linux Training
  vboxuser@kshitizgupta:~$ touch welcome.sh

- Prints current date
  vboxuser@kshitizgupta:~$ vi welcome.sh

- Prints current user name
  #!/bin/bash
  echo "Welcome to Linux Training"
  echo "Date: 15/03/26"
  echo "Name: Kshitiz Gupta"

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch welcome.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi welcome.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod +x welcome.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ./welcome.sh
Welcome to Linux Training
Date: 15/03/26
Name: Kshitiz Gupta
```

Task 2: User Input Script
Create a script that:

- Takes user name as input
- Prints: "Hello <name>, welcome to Linux"

```
vboxuser@kshitizgupta:~$ touch user.sh
vboxuser@kshitizgupta:~$ vi user.sh
vboxuser@kshitizgupta:~$ chmod +x user.sh
vboxuser@kshitizgupta:~$ ./user.sh
```

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Hello $name, welcome to Linux"
```

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi welcome.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch user.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi user.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod +x user.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ./user.sh
Enter your name:
Kshitiz
Hello Kshitiz, welcome to Linux
```

```
vboxuser@kshitizgupta: /media/sf_Downloads/linux_practice
#!/bin/bash
echo "Enter your name:"
read name
echo "Hello $name, welcome to Linux"
~
~
```

Task 3: Even or Odd Program

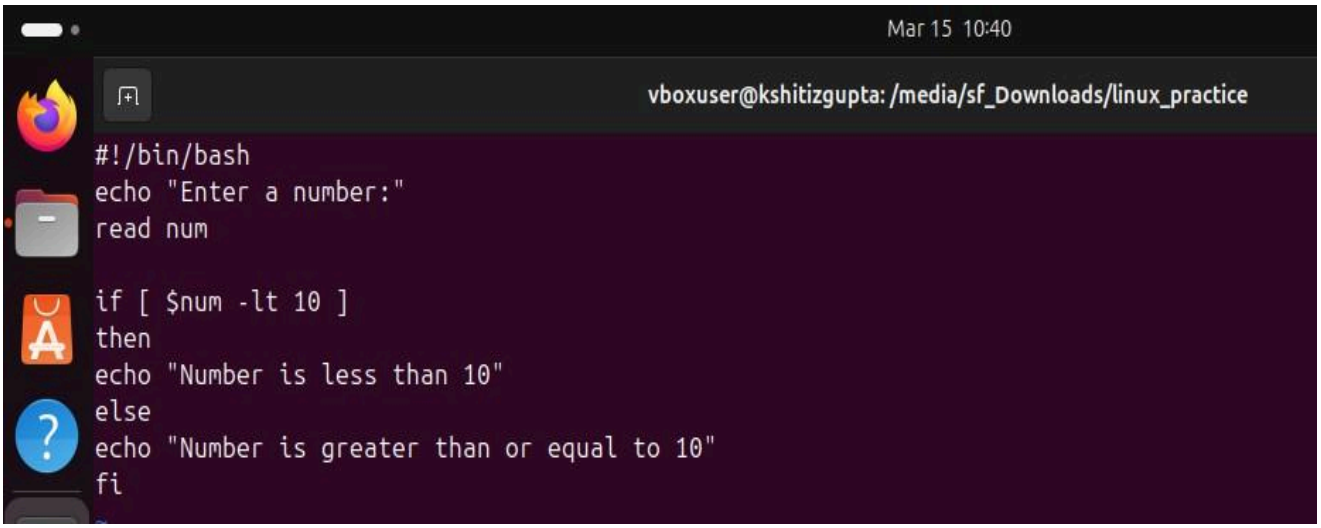
Write a script that:

- Takes a number as input
- Checks whether number is lesser than 10 or not

```
vboxuser@kshitizgupta:~$ touch number.sh
vboxuser@kshitizgupta:~$ vi number.sh
vboxuser@kshitizgupta:~$ chmod +x number.sh
vboxuser@kshitizgupta:~$ ./number.sh
```

```
#!/bin/bash
echo "Enter a number:"
read num

if [ $num -lt 10 ]
then
echo "Number is less than 10"
else
echo "Number is greater than or equal to 10"
fi
```



Task 4: Loop Program
Write a script that:
- Prints numbers from 1 to 10
- Prints multiplication table of 5

```
vboxuser@kshitizgupta:~$ touch loop.sh
vboxuser@kshitizgupta:~$ vi loop.sh
vboxuser@kshitizgupta:~$ chmod +x loop.sh
vboxuser@kshitizgupta:~$ ./loop.sh
```

```
#!/bin/bash
echo "Numbers from 1 to 10"
for i in {1..10}
do
echo $i
done

echo "Multiplication Table of 5"
for i in {1..10}
do
echo "5 x $i = $((5*i))"
done
```

vboxuser@kshitizgupta: /media/sf_Downloads/linux_practice

```
#!/bin/bash

echo "Numbers from 1 to 10"

for i in {1..10}
do
echo $i
done

echo "Table of 5"

for i in {1..10}
do
echo "5 x $i = $((5*i))"
done
~
~
```

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch loop.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi loop.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod +x loop.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ./loop.sh
Numbers from 1 to 10
1
2
3
4
5
6
7
8
9
10
Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Assignment 7: Mini Project

Create a script that:

1. Creates a directory backup
2. Takes current date and time
3. Copies all .txt files into backup folder
4. Prints "Backup Completed;
5. If backup folder already exists, print
6. "Directory already exists.

vboxuser@kshitizgupta:~$ touch backup.sh
vboxuser@kshitizgupta:~$ vi backup.sh
vboxuser@kshitizgupta:~$ touch backup.sh
vboxuser@kshitizgupta:~$ ./backup.sh

```
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ touch backup.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ vi backup.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ chmod +x backup.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ./backup.sh
Backup directory created
Current date and time:
Sun Mar 15 11:39:38 AM UTC 2026
Backup Completed
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls
backup  backup.sh  bio.txt  loop.sh  number.sh  script.sh  user.sh  welcome.sh
vboxuser@kshitizgupta:/media/sf_Downloads/linux_practice$ ls backup
bio.txt
```

vboxuser@kshitizgupta: /media/sf_Downloads/linux_practice

```
#!/bin/bash

if [ -d backup ]
then
echo "Directory already exists"
else
mkdir backup
echo "Backup directory created"
fi

echo "Current date and time:"
date

cp *.txt backup/

echo "Backup Completed"
~
```